

PHYSICS

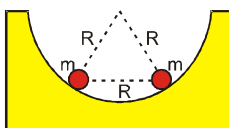
TARGET : JEE- Advanced

CAPS-11

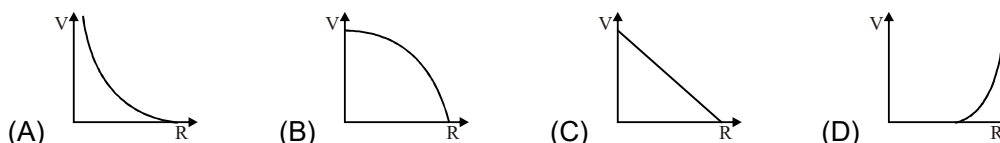
ELECTROSTATICS

SCQ (Single Correct Type) :

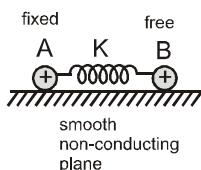
1. Two identical small balls each have a mass m and charge q . When placed in a hemispherical bowl of radius R with frictionless, nonconductive walls, the beads move, and at equilibrium the line joining the balls is horizontal and the distance between them is R (figure). Neglect any induced charge on the hemispherical bowl. Then the charge on each bead is: (here $K = \frac{1}{4\pi\epsilon_0}$)



- (A) $q = R \left(\frac{mg}{K\sqrt{3}} \right)^{1/2}$ (B) $q = \left(R \frac{mg}{K\sqrt{3}} \right)^{1/2}$ (C) $q = R \left(\frac{\sqrt{3} mg}{K} \right)^{1/2}$ (D) $q = \left(R \frac{\sqrt{3} mg}{K} \right)^{1/2}$
2. A small, charged pith ball moves on a smooth table toward an oppositely charged metal ball that is held in place on the table. Which of the curves shown in figure most realistically reflects the speed v of the pith ball as a function of the distance ' r ' between the balls? Initial distance between them is R .

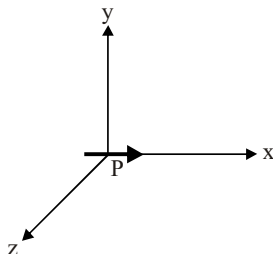


3. Two small identical balls lying on a horizontal plane are connected by a weightless spring (natural length ℓ_0). Ball A is fixed at O and the other ball B is free to move, with frequency f_0 . The balls are charged identically as a result of which the spring length increases by four times (new length is $4\ell_0$). Then new frequency is. (assume to ball as a point charge and amplitude of oscillation is very small)

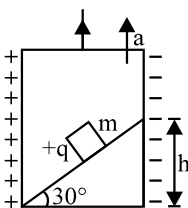


- (A) f_0 (B) $\frac{5}{2} f_0$ (C) $\sqrt{\frac{5}{3}} f_0$ (D) $\sqrt{\frac{5}{2}} f_0$

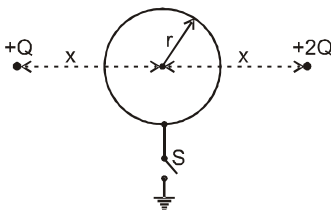
4. A short electric dipole is oriented along x-direction at origin. At which of following point the electric field have no x component.



- (A) $(1, 1, 0)$ (B) $(\sqrt{2}, 1, 0)$ (C) $(1, \sqrt{2}, 0)$ (D) $(1, 0, 0)$
5. A ball of mass m with charge q can rotate in a vertical plane at the end of a string of length ' ℓ ' in a uniform electrostatic field where lines of force are directed upwards. The minimum velocity which must be imparted to the ball at its lowest position so that string may not slaken at any position during the entire circular motion:
- (A) $\left[\frac{5\ell}{m}(mg - qE) \right]^{1/2}$ (B) $\left[\frac{5\ell}{m}(qE + mg) \right]^{1/2}$ (C) $\left[\frac{5\ell}{m}(mg + qE) \right]^{1/2}$ (D) None of these
6. A small block of mass ' m ' is kept at the top of a smooth inclined plane of angle 30° placed in an elevator moving upward with an acceleration ' a '. Electric field E exists between the vertical sides of the wall of the elevator. The charge on the block is $+q$. The time taken by the block to come to the lowest point of inclined plane is : [Take the surface to be smooth]

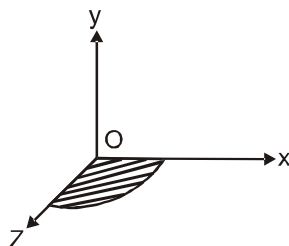


- (A) $\sqrt{\frac{2h}{g}}$ (B) $\sqrt{\frac{2h}{(g-a) + \frac{qE}{m}}}$ (C) $2\sqrt{\frac{2h}{(g+a) - \frac{qE}{m}\sqrt{3}}}$ (D) $\sqrt{\frac{2h}{(g+a)^2 - 3\left(\frac{qE}{m}\right)}}$
7. Two particles having positive charges $+Q$ and $+2Q$ are fixed at equal distance x from centre of an conducting sphere having zero net charge and radius r as shown. Initially the switch S is open. After the switch S is closed, the net charge flowing out of sphere is

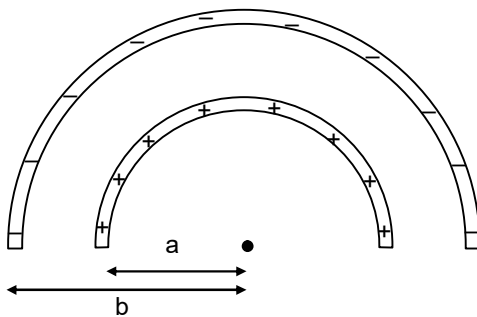


- (A) $\frac{Qr}{x}$ (B) $-\frac{Qr}{x}$ (C) $\frac{3Qr}{x}$ (D) $-\frac{3Qr}{x}$

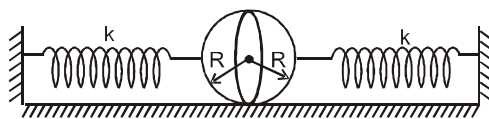
8. An infinitely long wire having uniform linear charge density λ is placed along positive y-axis such that one of its end point is at origin. Then the flux passing through quarter circular area (in x-z plane having centre at O) of radius R (as shown in figure) is :



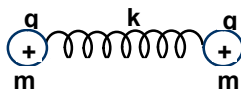
- (A) $\frac{\lambda R}{2\epsilon_0}$ (B) $\frac{\lambda R}{8\epsilon_0}$ (C) $\frac{\lambda R}{\epsilon_0}$ (D) $\frac{2\lambda R}{\epsilon_0}$
9. Figure shows two half coplanar rings of radii a and b. Both have same magnitude of charge but of opposite nature. Magnitude of charge on each ring is Q. What is electric dipole moment of the system ?



- (A) $\frac{2Q(b-a)}{\pi}$ (B) $\frac{Q(b-a)}{2\pi}$ (C) $\frac{Q(b+a)}{2\pi}$ (D) $\frac{4Q(b-a)}{3\pi}$
10. Two non-conducting hemispherical surfaces, which are having uniform charge density σ are placed on smooth horizontal surface as shown in figure. Assuming springs are ideal, calculate compression in each spring if both the hemispherical surface are just touching each other.

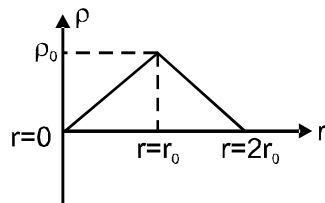


- (A) $\frac{\sigma^2 R^2}{2\epsilon_0 K}$ (B) R (C) $\frac{\sigma^2 \pi R^2}{2\epsilon_0 K}$ (D) $\frac{\sigma^2 \pi R^2}{2\epsilon_0 K}$
11. The ratio of the time periods of very small oscillation of the insulated spring and mass system before and after charging the masses is



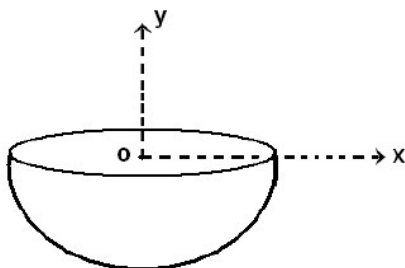
- (A) ≥ 1 (B) > 1 (C) ≤ 1 (D) $= 1$

12. Charge density (ρ) in a solid sphere varies with radial distance from centre (r) as shown in the graph :



Electric field intensity at a point $r = r_0$ is :

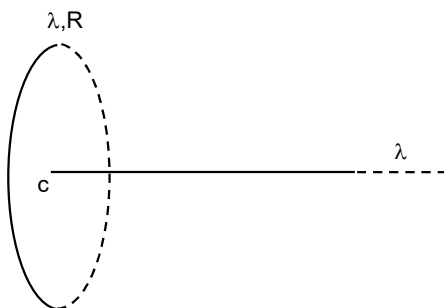
- (A) $\frac{\rho_0 r_0}{\epsilon_0}$ (B) $\frac{\rho_0 r_0}{2\epsilon_0}$ (C) $\frac{2\rho_0 r_0}{\epsilon_0}$ (D) $\frac{\rho_0 r_0}{4\epsilon_0}$
13. Consider a uniformly charged thin hemispherical shell of charge with charge density σ and radius R . A small freely oriented dipole with dipole moment $\vec{p}$ oriented along positive Y-axis lies at the center of the hemisphere. Moment of inertia of the dipole about an axis perpendicular to its length and passing through its center of mass is I .



The field intensity at the center of the hemisphere is

- (A) $\frac{\sigma}{\epsilon_0} \hat{j}$ (B) $\frac{\sigma}{2\epsilon_0} \hat{j}$ (C) $\frac{\sigma}{3\epsilon_0} \hat{j}$ (D) $\frac{\sigma}{4\epsilon_0} \hat{j}$
- MCQ (One or more than one correct) :**

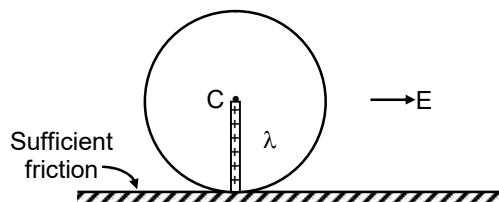
14. A semi-infinite wire of charge per unit length λ is placed along axis of uniformly charged ring of same charge density. One end of semi-infinite wire is at centre of the ring of the radius R . Select the correct statements :



- (A) Force on ring due to wire is $\frac{\lambda^2}{\epsilon_0}$ (B) Force on ring due to wire is $\frac{\lambda^2}{2\epsilon_0}$
- (C) Tension in ring due to wire is $\frac{\lambda^2}{4\pi\epsilon_0}$ (D) Tension in ring due to wire is zero

Comprehension Type Question:

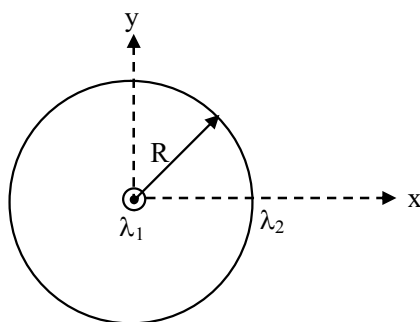
Consider a ring of mass m and radius R . A rod of mass m and length R is rigidly attached with ring along the radius as shown in the figure. The rod is uniformly charged with charge per unit length λ . There exist horizontal and uniform electric field E .



15. If ring rolls without slipping then Just after release :
- (A) Torque of electrostatics force about centre of ring is $\frac{\lambda ER^2}{2}$.
- (B) Torque of electrostatics force about bottom most point is $\frac{\lambda ER^2}{3}$.
- (C) Angular acceleration of rod is $\frac{3\lambda E}{14m}$.
- (D) Angular acceleration of rod is $\frac{\lambda E}{14m}$.
16. Find the angular velocity of system at the instant the rod reaches uppermost level for the first time.
- (A) $\omega = \sqrt{\frac{6(qE\pi - mg)}{13mR}}$ (B) $\omega = \sqrt{\frac{6(qE\pi + mg)}{13mR}}$ (C) $\omega = \sqrt{\frac{3(qE\pi - mg)}{13mR}}$ (D) $\omega = \sqrt{\frac{6(qE\pi - mg)}{7mR}}$

Numerical based Questions :

17. Consider a uniformly charged ring having linear charged density λ_2 and an infinite uniformly charged straight wire having linear charge density λ_1 , lying along the axis of ring as shown in figure. If force of interaction between wire and any half of the ring is found to be $F = 4k\lambda_1\lambda_2 R^{(\alpha-3)}$. Where α is an integer find α .

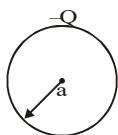


Matrix Match Type :

18. In each situation of column-I, some charge distributions are given with all details explained. The electrostatic potential energy and its nature is given situation in column-II.

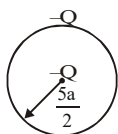
Column I**Column II**

- (A) A thin shell of radius a and having a charge $-Q$ uniformly distributed over its surface as shown



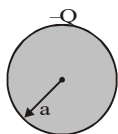
(p) $\frac{1}{8\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

- (B) A thin shell of radius $\frac{5a}{2}$ and having a charge $-Q$ uniformly distributed over its surface and a point charge $-Q$ placed at its centre as shown



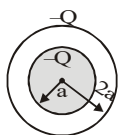
(q) $\frac{3}{20\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

- (C) A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume as shown



(r) $\frac{2}{5\pi\epsilon_0} \frac{Q^2}{a}$ in magnitude

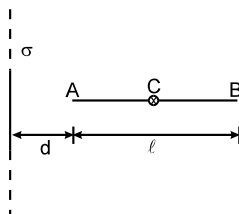
- (D) A solid sphere of radius a and having a charge $-Q$ uniformly distributed throughout its volume. The solid sphere is surrounded by a concentric thin uniformly charged spherical shell of radius $2a$ and carrying charge $-Q$ as shown



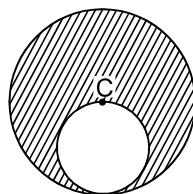
(s) Positive in sign

Subjective Type Questions :

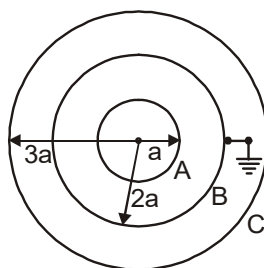
19. A uniform rod AB of mass m and length ℓ is hinged at its mid point C. The left half (AC) of the rod has linear charge density $-\lambda$ and the right half (CB) has $+\lambda$, where λ is constant. A large non conducting sheet of uniform surface charge density σ is also present near the rod. Initially, the rod is kept perpendicular to the sheet. The end A of the rod is initially at a distance d . Now the rod is rotated by a small angle in the plane of the paper and released. Prove that the rod will perform SHM and find its time period.



20. A solid sphere of radius 'R' is uniformly charged with charge density ρ in its volume. A spherical cavity of radius $\frac{R}{2}$ is made in the sphere as shown in the figure. Find the electric potential at the centre of the sphere.

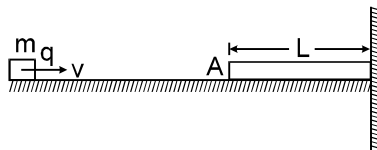


21. The potential difference between two large parallel plates is varied as $v = at$; a is a positive constant and t is time. An electron starts from rest at $t = 0$ from the plate which is at lower potential. If the distance between the plates is L , mass of electron m and charge on electron $-e$ then find the velocity of the electron when it reaches the other plate.
22. Figure shows a system of three concentric metal shells A, B and C with radii a , $2a$ and $3a$ respectively. Shell B is earthed and shell C is given a charge Q . Now if shell C is connected to shell A, then find the final charge on the shell B

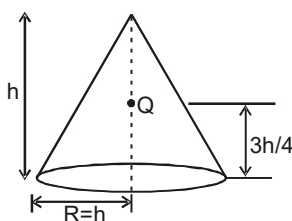


23. A non-conducting disc of radius a and uniform positive surface charge density σ is placed on the ground, with its axis vertical. A particle of mass m and positive charge q is dropped, along the axis of the disc, from a height H with zero initial velocity. The particle has $q/m = 4 \epsilon_0 g / \sigma$. Find the value of H if the particle just reaches the disc.

24. Figure shows a rod of length L which is uniformly charged with linear charge density λ kept on a smooth horizontal surface. Right end of rod is in contact with a vertical fixed wall.



- A block of mass m and charge q is projected with a velocity v from a point very far from rod in the line of rod. Find the distance of closest approach between the block & the left end A of the rod.
25. A charge Q is placed at a height $\frac{3h}{4}$ from the base of a cone of height h and radius R . Height and radius of cone are equal in magnitude. Calculate the flux linked with the slant curved surface of the cone.



ANSWER KEY (CAPS-11)

- | | | | | |
|---|------------------------|---|---|----------|
| 1. (A) | 2. (A) | 3. (D) | 4. (C) | 5. (A) |
| 6. (C) | 7. (C) | 8. (B) | 9. (A) | 10. (C) |
| 11. (D) | 12. (D) | 13. (D) | 14. (BC) | 15. (AC) |
| 16. (A) | 17. 3 | 18. (A) p,s (B) q,s (C) q,s (D) s | | |
| 19. $T = 2\pi \sqrt{\frac{2m\epsilon_0}{3\lambda\sigma}}$ | | 20. $V = \frac{5\rho R^2}{12\epsilon_0}$ | 21. $V = \left(\frac{9}{2} \frac{eaL}{m}\right)^{1/3}$ | |
| 22. $-\frac{8Q}{11}$ | 23. $H = \frac{4}{3}a$ | 24. $r = \frac{L}{\frac{2\pi\epsilon_0 mv^2}{\lambda q} - 1}$ | 25. $\frac{Q}{\epsilon_0} - \frac{Q}{5\epsilon_0} = \frac{4Q}{5\epsilon_0}$ | |